Lifestyle and Participant Factors for 30-Minute Blood Glucose Forecasting on CGMacros

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1 Introduction

This project studies whether lifestyle and participant information improves 30-minute glucose forecasting beyond recent glucose history. It uses CGMacros [2], which contains multimodal free-living data from 45 participants over approximately ten days, including continuous glucose measurements, diet and activity records, and participant-level information.

Two established time-series baselines are reproduced and adapted to this task, alongside a controlled factor-ablation study using recent glucose history as the reference feature set. Activity, Diet, and Bio are evaluated under both within-subject future prediction and unseen-subject generalization, with attention to prediction-time availability and uncertainty.

Glucose history provides a strong forecasting reference; Diet provides a small benefit in the held-out unseen-subject evaluation used here, while the other core factor groups do not show consistent transferable gains.

2 Related Work

Two established lines of work motivate the formally reproduced baselines used here. Sparacino et al. [5] proposed a first-order autoregressive model with a time-varying coefficient, re-estimated at every time point by weighted least squares, for glucose prediction 30 and 45 minutes ahead in people with type 1 diabetes; the same group later compared related algorithms directly [6]. We adopt the Sparacino et al. [5] model as a secondary formally reproduced baseline. More recently, Sergazinov et al. [4] introduced a curated benchmark of CGM datasets with a common evaluation protocol, including a simple global linear-regression model on past target lags that is reported to be competitive with substantially more complex sequence models; we adopt this model's definition as our primary adapted published baseline for the 120-minute-history CGMacros task.

A separate line of work predicts personalized postprandial glycemic response (PPGR) to specific meals rather than continuous glucose trajectories: Zeevi et al. [7] introduced machine-learning prediction of PPGR from meal composition and clinical variables, and Mendes-Soares et al. [3] showed a similar model generalizes across cohorts. The CGMacros dataset paper's own validation analysis follows this line of work, predicting breakfast PPGR area under the curve from anthropometric, biochemical, and macronutrient predictors [2]; Section 3 defines how this differs from the continuous forecasting task addressed here, and Appendix C revisits the PPGR analysis directly as a supplementary reconstruction.

3 Dataset and Prediction Task

CGMacros records approximately ten days from 45 adults: 15 without diabetes, 16 with pre-diabetes, and 14 with type 2 diabetes [2]. It combines Abbott FreeStyle Libre Pro and Dexcom G6 Pro CGMs (native sampling intervals of 15 and 5 minutes), Fitbit activity signals, meal logs and photographs, demographic and laboratory measurements, and partially available gut-health and microbiome data. A subset of participants additionally provided a stool sample analyzed for gut-microbiome composition and derived gut-health scores. Although collected in a free-living setting, breakfast and lunch macronutrient compositions followed study-designed menus, while dinners were freely chosen. Libre is the primary history signal and target because it is more complete in the released rows (99.97% versus 91.6% for Dexcom).

3.1 Forecasting Windows

At each origin t, the task predicts Libre glucose at $t + 30$ minutes using the closed history interval $[t - 120, t]$. Origins are spaced five minutes apart, anchored to each participant's first released timestamp. A window requires non-missing released glucose throughout its history and at its exact target time, with no gap in released timestamps anywhere in $[t - 120, t + 30]$. These fixed rules admit 134,539 of 141,263 candidate windows (95.2%) before protocol-specific splitting; they exclude incomplete record boundaries, missing values, and gap-crossing windows.

3.2 Relationship to the CGMacros PPGR Task

The original paper predicts breakfast two-hour area under the curve (AUC) and incremental AUC (iAUC) from anthropometric, biochemical, macronutrient, and meal-start glucose predictors, using leave-one-participant-out evaluation [2]. Here the target is a continuously updated future glucose value, and additional factors are assessed relative to this project's glucose-history reference. To be specific, given a participant's glucose history up to and including time t, we aim to forecast that participant's Libre glucose value at $t + 30$ minutes. Diet and core Bio variables were already represented in the paper's PPGR predictors; Activity was available in the dataset but absent from that stated predictor set. The separate PPGR partial reconstruction appears in Appendix C.

4 Data Processing and Feature Construction

4.1 Alignment and Prediction-Time Availability

The original CGMacros authors linearly interpolated the Libre device's native 15-minute readings and the Dexcom device's native 5-minute readings onto a common one-minute grid before release; individual native-reading anchors are not identified [2]. This project harmonizes schemas and aligns each participant's record to a one-minute index, preserving released values and inserting missing rows for absent timestamps. It performs no additional numerical interpolation or forward filling. Because the released one-minute series was interpolated by the dataset authors and native-reading anchors are unavailable, this study evaluates forecasting of the released offline signal rather than claiming deployable real-time prediction. Modeling uses unmerged meal events with normalized meal types; candidate merged episodes from the data audit are not used for features. Static data are joined by participant identifier.

Inputs are restricted to information at or before t, subject to explicit availability assumptions: an event timestamp alone does not establish when a device reading or log arrived. Meal type and nutrients are assumed available at their logged timestamp. Meal entry and upload times are

unrecorded, so this is an availability assumption; sensitivity analysis S1 delays Diet features by 30 minutes. Photographs and final consumed proportions are excluded because their availability at t is unverified.

4.2 Factor Discovery and Feature Groups

Candidate factors were drawn from training-scope exploration and data audit, then screened for prediction-time availability, coverage, and data-quality ambiguity before ablation. Table 1 summarizes how the available modalities were assigned to core, supplementary, exploratory, or excluded roles. Descriptive associations were not treated as evidence of predictive improvement, causal importance, or physiological ranking; predictive value was evaluated later under the locked ablation framework. The four core feature groups are defined below.

Table 1: Training-scope factor discovery and prediction-time roles.

Candidate source	Training-scope / availability consideration	Final role
Recent glucose	Dense primary history signal; available at prediction time under the task definition.	Core G
Heart rate / activity calories / MET	Time-varying activity; incomplete coverage is represented explicitly rather than coded as rest.	Core A
Meal logs / macronutrients	Recent dietary information; logged timestamps are used under an explicit availability assumption.	Core D
Age / gender / BMI	Static participant information, available for all participants.	Core B
Laboratory variables	Result-return timing is not established for online forecasting.	Offline S2 extension
Gut-health scores	Partial participant coverage.	Restricted-cohort sensitivity
Microbiome	Partial coverage and high-dimensional subject-level data.	Exploratory extension
Meal photographs	Prediction-time availability and reliable quantitative nutrient interpretation are not established.	Excluded
HbA1c	Unresolved dictionary/unit ambiguity in the forecasting pipeline.	Excluded from core forecasting; handled separately where justified in PPGR reconstruction

G**: glucose history.** Ten features: glucose at lags 0, 30, 60, and 120 minutes, plus mean, population standard deviation, and OLS slope with intercept over each of the closed 30- and 120-minute trailing windows.

A**: activity.** Over both 30- and 120-minute windows, heart rate, activity calories, and MET contribute observed-value means, maxima, coverage fractions, and missingness indicators; activity calories also contribute observed sums. MET and the alternative Intensity schema are not interchangeable. Missing activity is not coded as rest or zero.

D**: diet.** Over both 30- and 120-minute windows, logged-meal counts, no-logged-meal flags, and sums of calories, carbohydrate, protein, fat, and fiber are computed. The most recent admissible meal within 120 minutes supplies elapsed time, calories, carbohydrate, and meal-type

indicators. Nutrient totals are missing when no meal is logged or a contributing nutrient is missing, rather than assuming zero intake.

B**: core Bio.** Age, self-reported gender, and BMI, available for all 45 participants and assumed known after intake measurement. Laboratory values have unverified result-return times and are reserved for the offline S2 extension. HbA1c is excluded from the forecasting Bio configurations because of its unresolved unit ambiguity; its separate PPGR use is documented in Appendix C.

5 Methods

The two formal baselines reproduce published methods adapted to this task, rather than source-paper numerical scores. Simple references and extensible comparison models have separate roles.

5.1 Adapted Published Baselines

GlucoBench linear regression. A global OLS model with an intercept directly predicts the target from glucose lags at five-minute spacing [4]. Within the fixed 120-minute task history, candidate input lengths of 12 and 24 lags were compared using validation subject-equal RMSE; both protocols selected 24 lags for the locked test evaluation. No participant identifier or static covariate enters the model. This adaptation was checked against the public implementation.

Sparacino AR(1)-WLS. A local AR(1) coefficient is estimated at each origin by exponentially weighted least squares, with past pairs weighted by powers of a forgetting factor μ [5]. The 120-minute closed history window supplies 41 samples at three-minute spacing (40 consecutive regression pairs). With the fitted coefficient held fixed, future innovations are set to zero and the forecast is iterated for ten three-minute steps to produce the 30-minute-ahead point prediction. The project samples the released Libre grid, truncates fitting history to the task window, and omits the source study's low-pass filter. The candidates are $\mu \in \{0.2, 0.5, 0.8\}$, with validation subject-equal RMSE selecting $\mu = 0.2$ in both protocols. Closed-form WLS implements the published fitting objective; no reference implementation was available to establish numerical equivalence with the original RLS routine.

5.2 References and Factor-Ablation Models

Persistence repeats glucose at t; linear trend adds 30 times the trailing 30-minute slope to that value. They provide simple references and are not published baseline reimplementations.

Ridge is the primary ablation model because it accepts arbitrary combinations of G, A, D, B under a common fitting procedure. A shallow gradient-boosted-tree model (GBT) provides a model-family check for the G-only and full $G{+}A{+}D{+}B$ configurations. These models use the engineered G features, whereas GlucoBench uses its lag-vector specification and Sparacino uses its local univariate history.

Exact estimators, preprocessing, candidate grids, and validation-selected values are reported in Appendix E.

6 Experimental Design

6.1 Protocols and Splits

Protocol A: within-subject future prediction. Each participant's recorded time span is divided approximately 60/20/20 into training, validation, and test. The complete closed support $[t-120, t+30]$ must lie within one segment; boundary-straddling windows are excluded. Training includes its right boundary; validation and test are left-open and right-closed. The split is by elapsed time, not window count.

Protocol B: unseen-subject generalization. The 45 participants are randomly assigned as mutually exclusive subject-level groups to training, validation, and test in a 27/9/9 split using a fixed random seed of 42. The split is not stratified by metabolic status or glucose measurements. A held-out participant may supply their admissible history at prediction time, but contributes no data to fitting or preprocessing. Shared exploratory analysis uses only Protocol A training periods from Protocol B's 27 training participants. Models and learned preprocessing are otherwise fitted separately for each protocol. Table 2 gives the resulting split sizes.

The two protocols answer different questions: how well a model forecasts a known participant's near-term future, and how well it generalizes to a participant it has never seen. Results are reported separately for each rather than averaged together.

Table 2: Number of forecasting windows and participants in each split, by evaluation protocol.

	Windows			Participants		
Evaluation protocol	Train	Validation	Test	Train	Validation	Test
Protocol A	79,300	26,330	26,269	45	45	45
Protocol B	81,958	24,259	28,322	27	9	9

6.2 Ablations and Comparison Cohorts

The design crosses G with all subsets of $\{A, D, B\}$, yielding eight configurations. Each factor is assessed both by adding it to G and by comparing the full configuration with the corresponding group omitted; Diet also has the incremental $G{+}A \rightarrow G{+}A{+}D$ contrast. Configuration order is not an importance ranking.

Core comparisons use identical windows with at least one observed heart-rate or activity-calorie reading in their 120-minute history, even for configurations without A. The core test cohort contains 44 participants and 21,083 windows in Protocol A, and 9 participants and 26,861 windows in Protocol B. Baseline evaluation uses all admitted windows in Table 2. An all-windows ablation retains activity-missing windows with explicit missingness features; Gut sensitivity uses a further-restricted common cohort and its own refitted reference. Scores from different cohorts cannot be directly subtracted to estimate feature gains.

6.3 Selection, Locking, and Reproducibility

Learned imputation, scaling, and model parameters are fitted only on the relevant training split. Ridge uses training-fitted imputation and standardization; all hyperparameter choices use validation subject-equal RMSE. Every pre-specified configuration and sensitivity enters the locked test evaluation, including those with null or adverse validation outcomes. Test evaluation uses the stored

training-only fits without refitting on train plus validation or adjusting the registered comparisons after test inspection.

A fresh-kernel rerun of the core pipeline from the raw release reproduced the reported test metrics, paired confidence intervals, and practical-effect classifications.

The core ablations and pre-specified sensitivities, including the reduced-cohort Gut analysis, belong to the locked evaluation. Microbiome, device/horizon, sequence-model, and PPGR analyses are separate exploratory extensions; they do not select or confirm the core results.

6.4 Metrics, Paired Comparisons, and Uncertainty

Primary mean absolute error (MAE) and root mean square error (RMSE) (mg/dL) are calculated per participant and then averaged equally, so participants with more windows do not dominate. Secondary pooled-window metrics weight windows equally. Subject-equal metrics are treated as primary because they prevent participants with unusually long or densely sampled records from dominating the reported error, and because the two evaluation protocols are themselves defined at the participant level. Paired ΔRMSE is the mean per-participant difference, to minus from, on identical windows; negative values favor the added feature or alternative model.

The pre-specified practical-effect classification labels a contrast improved when $\Delta\text{RMSE} \leq -0.1$ mg/dL and more than half of participants improve, worse when $\Delta\text{RMSE} \geq 0.1$ mg/dL and fewer than half improve, negligible when $|\Delta\text{RMSE}| < 0.1$ mg/dL, and mixed otherwise. The corresponding codes are I (improved), W (worse), N (negligible), and M (mixed). The superscript denotes this pre-specified practical-effect classification rather than a statistical-significance decision. An individual improves when their paired ΔRMSE is negative. The threshold is a project reporting rule, not an established clinical threshold.

Confidence intervals are separate from the practical-effect classifications. A paired subject-cluster bootstrap resamples participants with replacement, retaining each participant's windows and both predictions together (2,000 resamples, random seed 42; 95% percentile intervals). It captures participant resampling uncertainty for fixed predictions, not variability from retraining. Small test cohorts, especially Protocol B, limit interval stability and coverage; they do not necessarily yield wide empirical intervals. Confidence intervals are descriptive across multiple pre-specified contrasts and are not adjusted for multiplicity.

7 Results

7.1 Baseline Performance

On the common all-admitted test windows, GlucoBench LR has the lowest subject-equal RMSE among the six models in both protocols, with Ridge close behind (Table 3). Both improve substantially over persistence. Sparacino AR(1)-WLS and linear trend fail to beat persistence, while shallow GBT falls between the linear models and persistence.

The all-window Ridge scores are not interchangeable with the restricted-cohort scores below. As a descriptive same-window check on the core cohort, GlucoBench LR versus G-only Ridge has pooled RMSE of 11.06 versus 11.44 mg/dL in A and 11.38 versus 11.52 mg/dL in B; this cross-model check is not a locked paired contrast. A post-hoc, descriptive comparison of frozen GlucoBench LR and full-factor Ridge C04 (G+A+D+B) on exactly matched core windows (44 participants and 21,083 windows in A; 9 participants and 26,861 windows in B) gives subject-equal RMSEs of 10.96 versus 11.29 mg/dL in A and 11.33 versus 11.12 mg/dL in B, respectively. The paired subject-level ΔRMSE (C04 minus GlucoBench) is +0.335 mg/dL (95% CI 0.173 to 0.509) in A

Table 3: Locked test-set performance (mg/dL) for the adapted published baselines, simple references, and project comparison models, evaluated on all admitted test windows.

	Subject-equal		Pooled-window	
Model	MAE	RMSE	MAE	RMSE
Protocol A: 45 participants, 26,269 test windows				
Adapted published baseline				
GlucoBench linear regression	7.39	10.83	7.32	10.92
Sparacino AR(1)-WLS	10.60	16.63	10.71	17.11
Simple reference				
Persistence	10.04	15.02	10.12	15.56
Linear trend (30 min)	13.04	19.26	13.22	19.83
Project comparison				
Ridge (glucose history)	7.69	11.19	7.63	11.30
Shallow GBT (glucose history)	9.05	13.06	9.07	13.36
Protocol B: 9 participants, 28,322 test windows				
Adapted published baseline				
GlucoBench linear regression	7.43	11.22	7.43	11.27
Sparacino AR(1)-WLS	10.55	17.93	10.56	18.10
Simple reference				
Persistence	9.92	15.27	9.89	15.51
Linear trend (30 min)	12.80	19.46	12.80	19.61
Project comparison				
Ridge (glucose history)	7.64	11.37	7.64	11.40
Shallow GBT (glucose history)	8.94	13.27	8.94	13.43

and −0.202 mg/dL (95% CI −0.693 to 0.154) in B; negative values favor C04, and the Protocol B interval spans zero. All prediction rows were matched one-to-one by the existing fold key, and the paired subject-level bootstrap used 2,000 resamples, random seed 42, and 95% percentile intervals on fixed predictions without retraining; this comparison is not a pre-specified locked contrast and has no practical-effect classification or statistical-significance conclusion.

7.2 Factor Contrasts and Protocol Dependence

The complete locked test performance for all eight core configurations is shown in Table 4; Figure 1 provides the paired-contrast overview across core and sensitivity analyses. Table 5 reports the core paired contrasts. The G-only reference has subject-equal RMSE of 11.30 mg/dL in A and 11.48 mg/dL in B.

Diet is the only core group classified as improved under the pre-specified practical-effect rule with an interval excluding zero, and only in Protocol B. Its three contrasts reduce RMSE by approximately 2.4–2.6% relative to each contrast's own reference configuration. In Protocol A, all core intervals include zero. Activity intervals include zero in both protocols. Bio's two Protocol B intervals exclude zero, but their reductions are below the practical-effect magnitude threshold and remain classified as negligible; this is a small nonzero effect, not evidence of no association.

The validation pattern does not carry uniformly to test. All three Diet contrasts were classified as improved under the practical-effect rule in Protocol A validation (approximately −0.12 to

−0.13 mg/dL), but none retained that classification on test. In Protocol B, only $G \rightarrow G{+}D$ was classified as improved in validation; all three Diet contrasts were classified as improved on test. Across all 42 locked contrasts, validation and test practical-effect classifications differ for 11 of 21 contrasts in A and 5 of 21 in B.

Given this pattern, the Diet result is reported narrowly: adding dietary information produced a small, confidence-interval-supported reduction in test RMSE for the unseen-subject generalization protocol, and did not produce a confirmed reduction for the within-subject future-prediction protocol, despite an unconfirmed improvement at the validation stage in the latter case.

Table 4: Locked test-set performance (mg/dL) for the eight core ablation configurations on the activity-availability-restricted common cohort.

		Subject-equal		Pooled-window	
Config	Feature groups	MAE	RMSE	MAE	RMSE
Protocol A: 44 participants, 21,083 test windows					
C1	G	7.82	11.30	7.74	11.44
C2	$G{+}A$	7.77	11.32	7.74	11.44
C3	$G{+}A{+}D$	7.72	11.29	7.68	11.40
C4	$G{+}A{+}D{+}B$	7.75	11.29	7.70	11.41
C5	$G{+}D$	7.74	11.26	7.68	11.40
C6	$G{+}B$	7.85	11.29	7.74	11.43
C7	$G{+}D{+}B$	7.78	11.25	7.68	11.40
C8	$G{+}A{+}B$	7.79	11.32	7.75	11.45
Protocol B: 9 participants, 26,861 test windows					
C1	G	7.70	11.48	7.71	11.52
C2	$G{+}A$	7.68	11.44	7.69	11.48
C3	$G{+}A{+}D$	7.49	11.17	7.51	11.22
C4	$G{+}A{+}D{+}B$	7.43	11.12	7.44	11.17
C5	$G{+}D$	7.49	11.17	7.50	11.22
C6	$G{+}B$	7.69	11.45	7.69	11.48
C7	$G{+}D{+}B$	7.48	11.14	7.49	11.19
C8	$G{+}A{+}B$	7.61	11.40	7.62	11.44

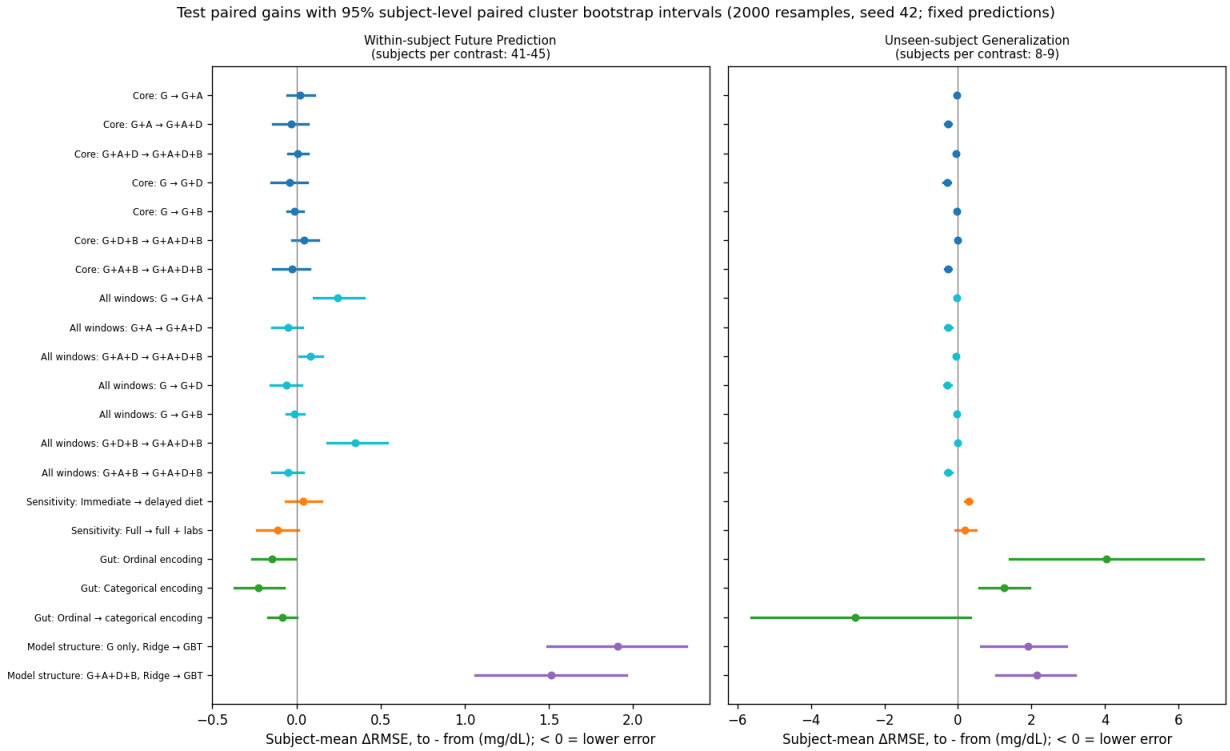


Figure 1: All 42 locked paired ΔRMSE contrasts with 95% subject-cluster bootstrap intervals. Negative values favor the destination configuration. The core block corresponds to Table 5; sensitivity, all-windows, and model-structure blocks extend Table 6; Gut uses its own reduced cohort (Appendix A.1). Horizontal scales differ between protocols.

Table 5: Core-cohort subject-mean paired ΔRMSE [95% CI], in mg/dL. Superscripts denote the pre-specified practical-effect classification: N = negligible, I = improved (Section 6.4); they are not statistical-significance decisions.

Contrast	Protocol A	Protocol B
Activity		
$G \rightarrow G{+}A$	$0.019\,[-0.056, 0.107]^{N}$	$-0.035\,[-0.070, 0.003]^{N}$
$G{+}D{+}B \rightarrow G{+}A{+}D{+}B$	$0.046\,[-0.028, 0.131]^{N}$	$-0.018\,[-0.063, 0.030]^{N}$
Diet		
$G{+}A \rightarrow G{+}A{+}D$	$-0.032\,[-0.142, 0.070]^{N}$	$-0.271\,[-0.363, -0.174]^{I}$
$G \rightarrow G{+}D$	$-0.041\,[-0.148, 0.062]^{N}$	$-0.301\,[-0.394, -0.206]^{I}$
$G{+}A{+}B \rightarrow G{+}A{+}D{+}B$	$-0.028\,[-0.140, 0.077]^{N}$	$-0.274\,[-0.369, -0.177]^{I}$
Bio		
$G{+}A{+}D \rightarrow G{+}A{+}D{+}B$	$0.005\,[-0.050, 0.068]^{N}$	$-0.045\,[-0.075, -0.022]^{N}$
$G \rightarrow G{+}B$	$-0.012\,[-0.056, 0.041]^{N}$	$-0.029\,[-0.055, -0.008]^{N}$

7.3 Robustness and Sensitivity Checks

Table 6 retains the locked checks that qualify the core findings. Delaying meal availability by 30 minutes worsens Protocol B performance by roughly the size of its immediate-Diet gain; the change in A is uncertain. Offline laboratory Bio has opposite point-estimate directions across protocols, but both intervals include zero, despite their practical-effect classifications as improved/worse.

On all admitted windows, both Activity contrasts worsen Protocol A performance with intervals excluding zero; Protocol B remains negligible. This differs from the core-cohort Activity result and cannot be generalized across cohorts. Shallow GBT underperforms Ridge in all four matched configuration/protocol comparisons, with every interval excluding zero. Gut sensitivity is reported separately on its reduced cohort in Appendix A.1; Figure 1 retains all 42 locked contrasts.

Table 6: Sensitivity and model-structure subject-mean paired ΔRMSE [95% CI], in mg/dL. Superscripts denote the pre-specified practical-effect classification: N = negligible, I = improved, W = worse; they are not statistical-significance decisions. For S2, both 95% CIs include zero; the I/W classifications are not confidence-interval-supported evidence of an effect in either direction.

Contrast	Protocol A	Protocol B
S1/S2: full feature set, core cohort		
Immediate → delayed Diet	$0.041\,[-0.063, 0.150]^{N}$	$0.284\,[0.186, 0.373]^{W}$
Full → full + labs	$-0.111\,[-0.234, 0.014]^{I}$	$0.186\,[-0.085, 0.492]^{W}$
Activity: all admitted windows		
$G \rightarrow G{+}A$	$0.243\,[0.102, 0.403]^{W}$	$-0.032\,[-0.064, 0.004]^{N}$
$G{+}D{+}B \rightarrow G{+}A{+}D{+}B$	$0.351\,[0.181, 0.539]^{W}$	$-0.014\,[-0.057, 0.032]^{N}$
Model structure: Ridge → shallow GBT, core cohort		
G only	$1.910\,[1.490, 2.319]^{W}$	$1.908\,[0.631, 2.959]^{W}$
$G{+}A{+}D{+}B$	$1.514\,[1.064, 1.960]^{W}$	$2.151\,[1.032, 3.195]^{W}$

8 Error Analysis

The following strata describe already-locked predictions post hoc. Realized glucose change, meals logged after the origin, and Dexcom at the target time use post-origin information and are diagnostic strata, not model inputs or confirmatory tests.

8.1 Error by Glucose Level and Realized Change

Figure 2 shows increasing error at high target glucose and during rapid changes. Ridge configurations behave similarly and generally improve on persistence, although persistence is competitive in the lowest glucose band. Per-participant error also varies with glucose variability, without establishing a feature-specific relationship.

8.2 Diet-Associated Error Stratification

Diet-associated reductions are concentrated in the first 30 minutes after a logged meal: for $G{+}D$ versus G, pooled ΔRMSE is -1.97 mg/dL in B and -0.72 mg/dL in A. Other meal-time strata show small or inconsistent differences. Meals logged only after the origin provide no gain; this localization is consistent with the S1 availability sensitivity.

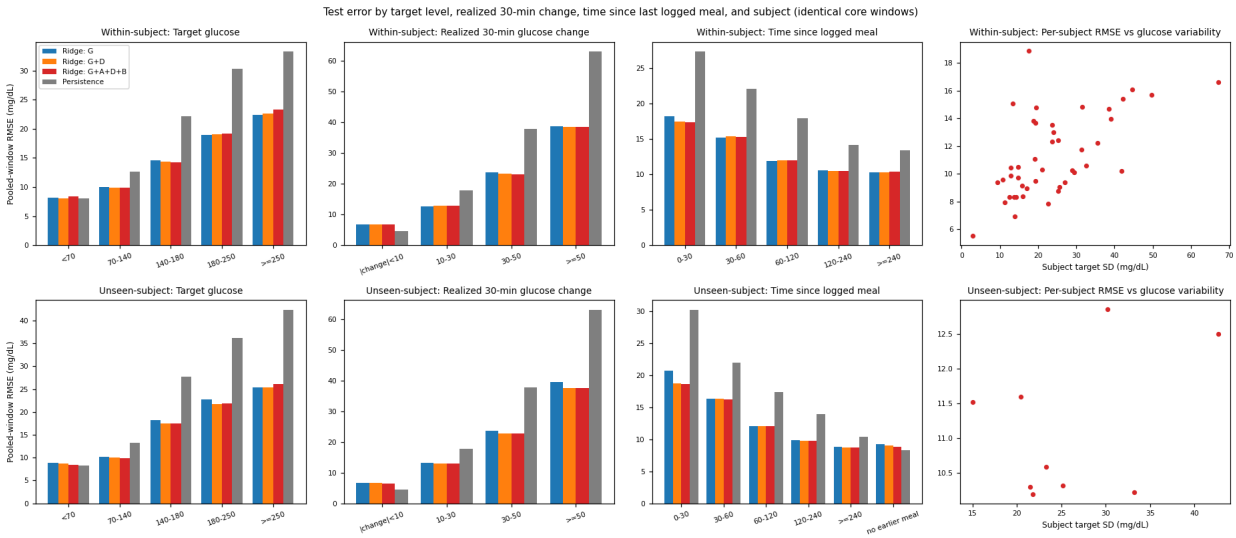


Figure 2: Pooled-window test RMSE stratified by target glucose level, realized 30-minute change, and time since the most recently logged meal, for persistence and three core-cohort Ridge configurations, on identical windows within each protocol. The rightmost panels show per-participant RMSE against that participant's own test-window glucose standard deviation.

8.3 Floor Censoring and Extreme Errors

The Libre record has a floor of 40 mg/dL. Across the two protocol evaluations, 2,130 test targets lie at this floor, concentrated in a few participants; these are evaluation counts, not independent observations across protocols. On core-cohort floor windows in B, Ridge and GlucoBench LR have positive bias of approximately 3.7–4.6 mg/dL against the recorded target. The latent glucose value below a censored reading is unobserved, so error against that latent value cannot be recovered or its direction inferred from the aggregate recorded-target bias.

Unconstrained linear predictions occasionally become negative in B (4 for each of the three core Ridge configurations shown, 11 for GlucoBench LR), whereas none do so in A; the models inspected in this diagnostic have no predictions above 400 mg/dL. Nine of the ten largest absolute-error cases in Protocol B occur in one participant and coincide with large Libre–Dexcom disagreements across several glucose-based models. The cause cannot be established as a sensor artifact or physiological event. In the bulk of windows, device disagreement does not show a consistent monotonic relationship with error, so the extreme episode is not a general device-error law.

9 Discussion and Limitations

9.1 Interpretation

Recent glucose history provides a strong reference for the 30-minute task. Within the locked Ridge ablation, additional factors do not provide a uniform benefit. Diet yields a small reduction in prediction error in the held-out unseen-subject evaluation used here, but not in within-subject future prediction, and this result depends on the assumed availability of meal logs. Activity and core Bio do not meet the pre-specified practical-effect classification. These findings should therefore be interpreted factor by factor and within the tested evaluation settings, rather than as a ranking of physiological importance.

The Diet finding concerns the incremental value of already logged meals. The test-set reduction in Protocol B is small in absolute terms (about 0.27–0.30 mg/dL) and is not confirmed in Protocol A; it depends on treating logged timestamps as availability times. The validation-to-test asymmetry is a warning that favorable validation contrasts do not reliably determine locked-test outcomes. Post-hoc error strata localize the Diet-associated reduction to the first 30 minutes after a logged meal, consistent with—but not proof of—the logged-time availability assumption tested by S1. Three Diet contrasts offer complementary comparisons on the same data, not independent replications. The within-Ridge factor gains should also be distinguished from cross-model performance: in the post-hoc same-window comparison, the full-factor Ridge model did not show a consistent advantage over GlucoBench LR across the two protocols.

Static measurements may partly encode participant-specific glucose offsets when the same participants appear in training and test. The locked Gut sensitivity's cross-protocol reversal is compatible with this interpretation, but does not establish an identity mechanism or transferable biological signal. Core Bio's small Protocol B reductions should likewise be distinguished from a practically substantial gain. Activity results depend on coverage and missing-data handling. These are predictive ablations under a particular representation and model family, not causal effects or a ranking of physiological importance; correlated features can substitute for one another.

9.2 Limits of Generalization

Participants and coverage. The study is small and short, with only nine unseen test participants and partially designed meals. Many overlapping windows do not create additional independent participants, and equal weighting can make participants with very few windows influential. Restricting every core configuration to activity-available windows improves comparability but limits the population of windows to which the results apply. Gut and microbiome analyses impose further coverage restrictions.

Measurement and availability. The interpolation described in Section 4 smooths short-lag glucose history and obscures which values are independent native measurements. Unknown interpolation/filter timing and sensor arrival latency prevent verifying that released inputs would have been available in real time; the same concern applies to the publisher's smoothed MET signal. The results therefore characterize offline forecasting of the released signal, and lag coefficients should not be interpreted as minute-resolution physiology. Complete-history admission also means accuracy during sensor gaps was not tested. Floor censoring and device disagreement further limit interpretation against latent glucose.

Meal upload times are unrecorded, and training-scope meal-response curves sometimes rise before the logged timestamp, suggesting imperfect alignment with eating onset. Laboratory, Gut, and microbiome report-return times are also unknown, so these extensions rely on offline availability assumptions. The HbA1c dictionary-unit conflict is unresolved for core forecasting; the PPGR reconstruction makes a separate explicit interpretation rather than changing that policy.

Models and evidence scope. Baseline adaptations change the device, sampling, and task context relative to the source studies. In particular, Sparacino's poor performance here cannot isolate the consequences of omitting filtering, using interpolated Libre samples, or changing the study population. Its three-minute-spaced samples come from the author-interpolated one-minute Libre grid, rather than the original study's native three-minute GlucoDay readings; this is an adaptation to a different device and sampling scheme, not a like-for-like replication. The Ridge penalty and GBT iteration choices often lie at their search-grid boundaries. Fixed tuning budgets support consistent within-family ablations but do not establish either family's best achievable performance. The shallow GBT result is limited to the matched tuning budget and does not rule out

other nonlinear methods. In the exploratory sequence-model checks, GRU and TCN had modestly lower test-RMSE point estimates than the adapted linear baseline, but their unseen-subject intervals included zero; these checks therefore do not establish a general advantage for nonlinear models either. The core claims remain limited to the primary task and tested representations; the later device/horizon and sequence-model checks in Appendix B are not independent confirmation. No feature-attribution analysis was used to separate correlated predictors. A check of consistency between meal photographs and logged meal information was considered, but post-meal or empty-plate images cannot be treated as standard evidence of a mismatch, and precise calories or nutrient quantities cannot be reliably inferred from images alone. A rigorous image–text consistency analysis would require annotation-supported calibration and held-out evaluation, which were not completed; this direction was therefore discontinued.

10 Conclusion

This project adapted two established forecasting baselines and evaluated lifestyle and participant factors through controlled ablations beyond glucose history. In the held-out unseen-subject evaluation used here, dietary information produced a small, availability-dependent reduction in prediction error, while other core factors offered no comparable gain and results varied across evaluation settings. This within-model gain did not establish a consistent advantage of the full-factor Ridge model over the GlucoBench baseline across the two evaluation protocols.

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A Gut-Health and Microbiome Analyses

A.1 Locked Gut-Health-Score Sensitivity

Ordinal and categorical encodings of 22 gut-health scores were compared with the full configuration refitted on the same Gut-available cohort: 41 participants in A and 8 in B. Ordinal encoding changes test subject-mean RMSE by −0.14 mg/dL (95% CI −0.27 to −0.01) in A and +4.04 (1.41 to 6.66) in B; categorical encoding changes it by −0.22 (−0.37 to −0.07) and +1.25 (0.58 to 1.96), respectively. Both encodings improve within-subject prediction and worsen unseen-subject prediction with intervals excluding zero, specifically on this restricted cohort.

A.2 Exploratory Microbiome Extension

A subject-level Ridge correction of full-model residuals uses binary microbial detection indicators. Screening and fitting use training participants; the regularization strength is selected on validation data. There are 44 participants with microbiome data available and 42 with Gut scores; these marginal availability counts are not the matched evaluation cohort. The microbiome–Gut matched test cohort contains 41 participants / 19,888 windows in A and 8 / 23,455 in B. Compared with the full model refitted on this cohort, test ΔRMSE is −0.20 mg/dL in A and −0.01 in B (validation: −0.17 and +0.02). This small within-subject change does not demonstrate transferable benefit for unseen participants.

B Exploratory Device, Horizon, and Sequence-Model Checks

B.1 Device and Horizon Sensitivity

Two tasks retain the history duration and model specification but use Dexcom history/target at 30 minutes or Libre at 60 minutes. GlucoBench LR beats same-task persistence in all four task/protocol comparisons (Table 7). Relative to primary-task test-window counts, retained coverage is 61.6%/88.3% for Dexcom and 98.9%/99.6% for Libre at 60 minutes (A/B). Absolute errors are higher, but device, horizon, coverage, and sample composition prevent interpreting these differences as a controlled change in model accuracy.

Table 7: Exploratory test-set performance (subject-equal MAE/RMSE, mg/dL) for GlucoBench linear regression under two supplementary task definitions.

Task	Protocol	MAE	RMSE	ΔRMSE*
Dexcom target, 30 min	A	11.94	16.65	−1.34
Dexcom target, 30 min	B	11.74	17.49	−2.11
Libre target, 60 min	A	14.47	20.39	−2.50
Libre target, 60 min	B	14.43	20.93	−2.95

*Relative to a same-task persistence reference (subject-equal RMSE); negative values indicate lower error than persistence.

B.2 Compact Sequence Models

A compact GRU (3,393 parameters) and causal TCN (2,465 parameters) were each trained with five fixed seeds on 25 glucose values spaced five minutes apart across the history, using validation

early stopping and the same test windows as the formal baselines. Four GRU seeds in A reached the epoch limit; the budget was not extended. GRU used GPU and TCN CPU, so their training times are not directly comparable.

Mean test subject-equal RMSE is 10.44/10.73 mg/dL for GRU and 10.45/10.72 for TCN (A/B). Paired differences versus GlucoBench LR are −0.40 (95% CI −0.61 to −0.21) and −0.38 (−0.62 to −0.17) in A, and −0.49 (−1.43 to 0.13) and −0.51 (−1.42 to 0.13) in B, for GRU and TCN respectively. Direct TCN-minus-GRU differences are 0.02 (−0.05 to 0.08) in A and −0.02 (−0.08 to 0.04) in B. The architectures give compatible exploratory results, with no supported winner between them.

C Original-Paper PPGR Partial Reconstruction

C.1 Published experiment

The original paper reports a breakfast-level task predicting the 2-hour area under the curve (AUC) and incremental area under the curve (iAUC) of the Libre response, using anthropometric, biochemical, and macronutrient predictors together with the glucose value at the start of the meal; an XGBoost model [1] with 80 trees, maximum depth 1, learning rate 0.2, L1 regularization of 1, and no L2 regularization; and a leave-one-participant-out evaluation with reference Pearson correlations of 0.89 (AUC) and 0.64 (iAUC). The paper does not fully specify the exact breakfast-inclusion rule, the treatment of missing or dictionary-flagged biomarker values, the HOMA-IR formula, the unit convention for the iAUC integral, or the random seed and software environment.

C.2 Public author-code implementation

The public code supplies operational choices not specified in the paper, including a HOMA-IR formula, AUC/iAUC integration details, a meal-start baseline, and train-only predictor scaling. It also uses positional row selection for meal windows without verifying timestamp continuity, aligns participant-level predictors by row position, and applies iAUC>0 and first-breakfast exclusions that are not stated as paper criteria. Its nutrient conversion applies additional scaling beyond the reconstruction's single conversion. These code behaviors are evidence about that implementation, not authoritative specifications of the published experiment.

C.3 Evidence-bound reconstruction

The reconstruction treats one eligible breakfast as one sample and predicts AUC and iAUC separately. It uses gap-free, timestamp-verified two-hour windows and trapezoidal integration in mg/dL·min, with iAUC calculated as positive area above the meal-start baseline. The 19 predictors are carbohydrate, protein, fat, and fiber energy; meal-start glucose; age, gender, BMI, HbA1c, HOMA-IR, fasting insulin, triglycerides, cholesterol, HDL, non-HDL, LDL, CHO/HDL ratio, VLDL, and fasting glucose. Subject identity is used only to join participant-level data and to define leave-one-participant-out folds. HOMA-IR is derived as fasting insulin times fasting glucose divided by 405. HbA1c is used on an explicit percent-scale interpretation because the dataset dictionary's mmol/mol label conflicts with the supplied values. Dictionary-designated error values are masked and complete cases are used. Where the paper and public code leave choices unresolved, these are stated reconstruction assumptions rather than claims about the original experiment.

The paper, public author code, and the reconstruction do not fully coincide; where the public code makes undocumented positional or scaling choices, these are treated as implementation evi-

dence rather than as authoritative specifications of the published experiment. Accordingly, this is an evidence-bound partial reconstruction, not an exact reproduction.

C.4 Primary result and sensitivity

On the complete-case primary cohort of 409 breakfasts from 43 participants, pooled Pearson correlations are 0.845 for AUC and 0.601 for iAUC (both $p < 0.001$), compared descriptively with the paper's 0.89 and 0.64. This numerical difference is not attributed to a single implementation choice. Applying the public code's iAUC> 0 restriction leaves 403 meals and changes the AUC/iAUC correlations to 0.867/0.602. Applying its additional nutrient scaling leaves the predictions unchanged. A positional join of static predictors would assign records to the wrong participant for 37 of the 43 participants in this release; the reconstruction therefore joins by participant ID. This diagnostic identifies a risk in positional assignment and does not imply that the published analysis used an incorrect assignment.

D Supplementary Error Trajectories

Figure 3 shows the participants with the lowest, median, and highest full-model Ridge subject-level RMSE on the core test cohort for each protocol, with each participant represented by the test day containing the most core-cohort test windows. These selected trajectories provide qualitative visual evidence for the quantitative error patterns discussed in Section 8, illustrating model behavior and limitations associated with floor censoring, rapid glucose changes, and meal timing. They are not a statistical summary of the cohort or confirmatory evidence, and the selected participants and days cannot be generalized to the cohort.

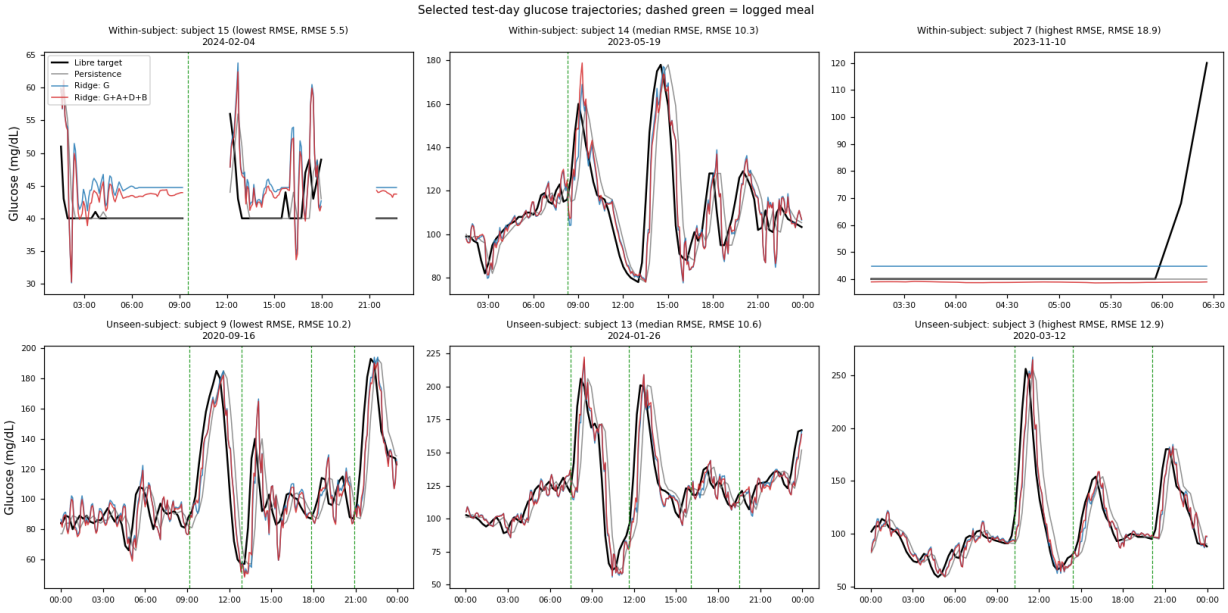


Figure 3: Test-day glucose trajectories for the participants with the lowest, median, and highest full-model Ridge subject-level RMSE on the core test cohort in each protocol. The shown day is the participant's test day with the most core-cohort test windows.

E Model and Tuning Specification

The factor-ablation pipeline fits one model for each registered configuration, cohort, evaluation protocol, and model family. Hyperparameters are selected by the lowest validation subject-mean RMSE; test data do not enter selection. The configuration identifiers below match the locked artifact registry (C01–C08 correspond to C1–C8 in the report).

E.1 Estimators, preprocessing, and candidate grids

Ridge uses `sklearn.linear_model.Ridge`, searched over $\alpha \in \{0.1, 1.0, 10.0\}$. The project pipeline first applies `SimpleImputer(strategy="median", keep_empty_features=True)` and then `StandardScaler`; both are fitted on training rows only. The core static inputs use their registered representations: raw numeric age and BMI and a male indicator with female as the reference level; no additional categorical encoding step is applied in the Ridge pipeline.

The GBT uses `sklearn.ensemble.HistGradientBoostingRegressor`, searched over the two combinations $(\texttt{learning_rate}, \texttt{max_depth}, \texttt{max_iter}) = (0.1, 3, 100)$ and $(0.1, 3, 300)$. The implementation sets `random_state=0` and `early_stopping=False`; it passes engineered inputs directly to the estimator without a project-level imputation or scaling pipeline.

E.2 Validation-selected values

Table 8 reports the selected hyperparameters recorded for each protocol and configuration. Configurations grouped in a row share the stated value; the GBT C08 Protocol B choice differs by cohort.

Table 8: Validation-selected factor-ablation hyperparameters.

Model	Protocol	Configuration / cohort	Selected value
Ridge	A	C01–C08, S01, S02, S03 categorical and ordinal	$\alpha = 0.1$
Ridge	B	C01–C08, S01, S02, S03 categorical	$\alpha = 0.1$
Ridge	B	S03 ordinal (Gut cohort)	$\alpha = 10.0$
GBT	A	C01–C08, S01, S02, S03 categorical and ordinal	`max_iter` = 300
GBT	B	C01–C03, C05	`max_iter` = 300
GBT	B	C04, C06–C07, S01–S02, S03 categorical and ordinal	`max_iter` = 100
GBT	B	C08, core cohort	`max_iter` = 100
GBT	B	C08, full cohort	`max_iter` = 300

For every GBT row, the selected values of `learning_rate` and `max_depth` are 0.1 and 3, respectively. The values shown are those in the saved validation fits; no additional tuning was performed for this report. Table 8 records all configurations fitted during model-family screening, whereas the locked GBT test comparisons retain only C01 and C04, as specified in Section 5.2.